

What is the cell hydration status of healthy children in the USA? Preliminary data on urine osmolality and water intake.

OBJECTIVE: Hyperosmotic stress on cells limits many aspects of cell function, metabolism and health. International data suggest that schoolchildren may be at risk of hyperosmotic stress on cells because of suboptimal water intake. The present study explored the cell hydration status of two samples of children in the USA. **DESIGN:** Cross-sectional study describing the urine osmolality (an index of hyperosmotic cell shrinkage) and water intake of convenience samples from Los Angeles (LA) and New York City (NYC). **SETTING:** Each participant collected a urine sample at an outpatient clinic on the way to school on a weekday morning in spring 2009. Each was instructed to wake, eat, drink and do as usual before school, and complete a dietary record form describing the type and amounts of all foods and beverages consumed after waking, before giving the sample. **SUBJECTS:** The children (9-11 years) in LA (n 337) and NYC (n 211) considered themselves healthy enough to go to school on the day they gave the urine sample. **RESULTS:** Elevated urine osmolality (>800 mmol/kg) was observed in 63 % and 66 % of participants in LA and NYC, respectively. In multivariable-adjusted logistic regression models, elevated urine osmolality was associated with not reporting intake of drinking water in the morning (LA: OR = 2.1, 95 % CI 1.2, 3.5; NYC: OR = 1.8, 95 % CI 1.0, 3.5). Although over 90 % of both samples had breakfast before giving the urine sample, 75 % did not drink water. **CONCLUSIONS:** Research is warranted to confirm these results and pursue their potential health implications.